Special Factoring: Perfect Squares, Difference of Squares & Grouping

Notes · Summary · Practice Problems · Full Answer Key

Name: _____ Date: _____

Estimated time: 45–60 minutes

1. Summary

Pattern A — Perfect Square Trinomial.

$$a^2 + 2ab + b^2 = (a+b)^2 \qquad \text{and} \qquad a^2 - 2ab + b^2 = (a-b)^2.$$

The first and last terms are perfect squares (a^2, b^2) and the middle term is $2ab$.

Pattern B — Difference of Squares.

$$a^2 - b^2 = (a-b)(a+b).$$

Two perfect-square terms separated by a minus sign, with no middle term.

Pattern C — Factoring by Grouping (four terms).

(1) Group the four terms into two pairs.

(2) Factor out the GCF of the *first* two terms and the GCF of the *last* two terms.

(3) If both pairs now share the *same* binomial factor, factor it out.

$$ax + ay + bx + by = a(x+y) + b(x+y) = (x+y)(a+b).$$

Always check whether a resulting factor can be factored further (e.g. a difference of squares).

$$\boxed{(a+b)^2 = a^2 + 2ab + b^2} \quad \boxed{a^2 - b^2 = (a-b)(a+b)} \quad \boxed{ax + ay + bx + by = (x+y)(a+b)}$$

2. Worked Examples

Part A — Perfect Square Trinomials

A1. $x^2 - 12x + 36$: product 36, sum $-12 \Rightarrow -6, -6$.

$$x^2 - 12x + 36 = x^2 - 6x - 6x + 36 = (x-6)^2.$$

	x	-6
x	x^2	$-6x$
-6	$-6x$	36

A2. $x^2 + 6x + 9 \Rightarrow 3, 3$: $x^2 + 6x + 9 = (x+3)^2$. **A3.** $x^2 - 14x + 49 \Rightarrow -7, -7$: $x^2 - 14x + 49 = (x-7)^2$.

Part B — Difference of Squares

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B1. $x^2 - 9 = x^2 + 0 \cdot x - 9$; product -9, sum $0 \Rightarrow 3, -3$.

$$x^2 - 9 = x^2 + 3x - 3x - 9 = (x - 3)(x + 3).$$

	x	-3
x	x^2	$-3x$
3	$3x$	-9

B2. $x^2 - 100 = (x + 10)(x - 10)$.

Part C — Factoring by Grouping

C. Factor $2z^3 + 6z^2 + 3z + 9$.

$$\begin{aligned} 2z^3 + 6z^2 + 3z + 9 &= (2z^3 + 6z^2) + (3z + 9) && \text{group into pairs} \\ &= 2z^2(z + 3) + 3(z + 3) && \text{GCF of each pair} \\ &= (z + 3)(2z^2 + 3) && \text{factor out the common } (z + 3) \end{aligned}$$

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3. Practice Problems

Part A — Factor each perfect square trinomial.

A1. $x^2 + 8x + 16$

A2. $x^2 - 10x + 25$

A3. $x^2 + 20x + 100$

A4. $x^2 - 2x + 1$

A5. $4x^2 + 12x + 9$

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A6. $9x^2 - 30x + 25$

A7. $x^2 + x + \frac{1}{4}$

A8. $16x^2 - 40x + 25$

A9. $25x^2 + 60x + 36$

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A10. Application. The area of a square patio is $16m^2 - 72m + 81$ ft^2. Find one side.

Part B — Factor each difference of squares.

B1. $x^2 - 25$

B2. $x^2 - 49$

B3. $x^2 - 1$

B4. $4x^2 - 9$

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B5. $9x^2 - 16$

B6. $25x^2 - 64$

B7. $x^2 - 144$

B8. $49x^2 - 100$

B9. $x^2 - \frac{4}{9}$

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B10. Application. A square of side x has a square of side 8 cut out. Write the remaining area factored.

Part C — Factoring by Grouping.

For C1–C4: find the GCF of the first two terms and the GCF of the last two terms, then factor.

C1. $3a^3 + 12a^2 + 2a + 8$

C2. $6b^3 - 10b^2 + 9b - 15$

C3. $4c^3 + 20c^2 - 3c - 15$

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C4. $10d^3 + 4d^2 + 15d + 6$

For C5–C13: factor each expression completely.

C5. $7n^3 + 21n^2 + 2n + 6$

C6. $5t^3 - 15t^2 + 4t - 12$

C7. $12x^3 + 8x^2 + 15x + 10$

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C8. $9u^3 - 6u^2 + 15u - 10$

C9. $8s^3 + 12s^2 - 10s - 15$

C10. $6j^3 - 9j^2 - 8j + 12$

C11. $2e^3 + 6e^2 - 5e - 15$

C12. $4f^3 + 8f^2 - 9f - 18$

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C13. $25g^3 - 50g^2 - g + 2$

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4. Detailed Answer Key

Part A — Perfect Square Trinomials

A1. $a = x,\ b = 4;\ 2ab = 8x$ ✓. $\boxed{(x+4)^2}$

A2. $a = x,\ b = 5;$ middle $-10x$ ✓. $\boxed{(x-5)^2}$

A3. $a = x,\ b = 10;\ 2ab = 20x$ ✓. $\boxed{(x+10)^2}$

A4. $a = x,\ b = 1;$ middle $-2x$ ✓. $\boxed{(x-1)^2}$

A5. $a = 2x,\ b = 3;\ 2ab = 12x$ ✓. $\boxed{(2x+3)^2}$

A6. $a = 3x,\ b = 5;$ middle $-30x$ ✓. $\boxed{(3x-5)^2}$

A7. $a = x,\ b = \frac{1}{2};\ 2ab = x$ ✓. $\boxed{\left(x+\frac{1}{2}\right)^2}$

A8. $a = 4x,\ b = 5;$ middle $-40x$ ✓. $\boxed{(4x-5)^2}$

A9. $a = 5x,\ b = 6;\ 2ab = 60x$ ✓. $\boxed{(5x+6)^2}$

A10. $16m^2 - 72m + 81 = (4m-9)^2$, so side = $\boxed{4m - 9 \text{ ft}}$.

Part B — Difference of Squares

B1. $\boxed{(x-5)(x+5)}$

B2. $\boxed{(x-7)(x+7)}$

B3. $\boxed{(x-1)(x+1)}$

B4. $\boxed{(2x-3)(2x+3)}$

B5. $\boxed{(3x-4)(3x+4)}$

B6. $\boxed{(5x-8)(5x+8)}$

B7. $\boxed{(x-12)(x+12)}$

B8. $\boxed{(7x-10)(7x+10)}$

B9. $\boxed{\left(x-\frac{2}{3}\right)\left(x+\frac{2}{3}\right)}$

B10. Remaining area $= x^2 - 64 = \boxed{(x-8)(x+8)}$.

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Part C — Factoring by Grouping

Shown: GCF of first pair, GCF of second pair, then the common binomial.

C1. $3a^2(a+4)+2(a+4)=\boxed{(a+4)(3a^2+2)}$

C2. $2b^2(3b-5)+3(3b-5)=\boxed{(3b-5)(2b^2+3)}$

C3. $4c^2(c+5)-3(c+5)=\boxed{(c+5)(4c^2-3)}$

C4. $2d^2(5d+2)+3(5d+2)=\boxed{(5d+2)(2d^2+3)}$

C5. $7n^2(n+3)+2(n+3)=\boxed{(n+3)(7n^2+2)}$

C6. $5t^2(t-3)+4(t-3)=\boxed{(t-3)(5t^2+4)}$

C7. $4x^2(3x+2)+5(3x+2)=\boxed{(3x+2)(4x^2+5)}$

C8. $3u^2(3u-2)+5(3u-2)=\boxed{(3u-2)(3u^2+5)}$

C9. $4s^2(2s+3)-5(2s+3)=\boxed{(2s+3)(4s^2-5)}$

C10. $3j^2(2j-3)-4(2j-3)=\boxed{(2j-3)(3j^2-4)}$

C11. $2e^2(e+3)-5(e+3)=\boxed{(e+3)(2e^2-5)}$

C12. $4f^2(f+2)-9(f+2)=(f+2)(4f^2-9)=\boxed{(f+2)(2f-3)(2f+3)}$ (diff. of squares!)

C13. $25g^2(g-2)-1(g-2)=(g-2)(25g^2-1)=\boxed{(g-2)(5g-1)(5g+1)}$ (diff. of squares!)

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